



Operating

Given a detailed customer environment scenario including a specific request, explain the implications of incorporating the request directly in a production environment.

Develop your career with Salesforce Training and Certification Preparation

Table of Contents

-  [Direct Changes in Production](#)
-  [Business Scenario](#)
-  [Requirements & Solutions](#)



After studying this topic, you should be able to:

-  Identify the considerations for making changes directly in a production environment.
-  Describe the implications of making direct changes in production.
-  Describe the strategies that can be utilized for making direct changes in production.
-  Recommend suitable approaches for making direct changes in production for a company's requirements.



Introduction

Organizations leveraging Salesforce must navigate the delicate balance of implementing **urgent updates in their production environment** against the backdrop of **ensuring operational stability and security compliance**. It is important to understand the **critical considerations and implications** of directly incorporating requests in a production environment.

Appropriate **strategies** must be adopted for addressing immediate issues in Salesforce-driven operations. Salesforce architects need to make informed decisions about making changes directly in production. Urgent fixes should enhance system functionality without compromising data integrity or user experience. Furthermore, **best practices** should be considered for managing direct changes in production, emphasizing the importance of thorough testing, stakeholder communication, and compliance with industry standards.



Incorporating Requests Directly in Production

Changes in Sandbox

A sandbox is a risk-free testing environment that should be used for changes that require extensive testing or could impact system-wide operations, such as new workflow steps.

Implications

Implications of direct changes in production include the possibility of errors, conflicts with other concurrent updates or integrations, and regulatory and compliance risks.



Changes in Production

Direct changes in production should be approached with caution by utilizing steps such as double-checking, clear communication, phased release, and training.

Strategies

Certain strategies can be utilized for direct changes in production, including a dedicated process for hotfixes, review and authorization, and using release management tools.

Direct Changes in Production

Table of Contents



Direct Changes in Production

Organizations typically have a **production org with live data** and **one or more sandbox environments** for development and testing, ensuring **changes do not affect live data** or **disrupt business operations**.

❖ RISK-FREE TESTING

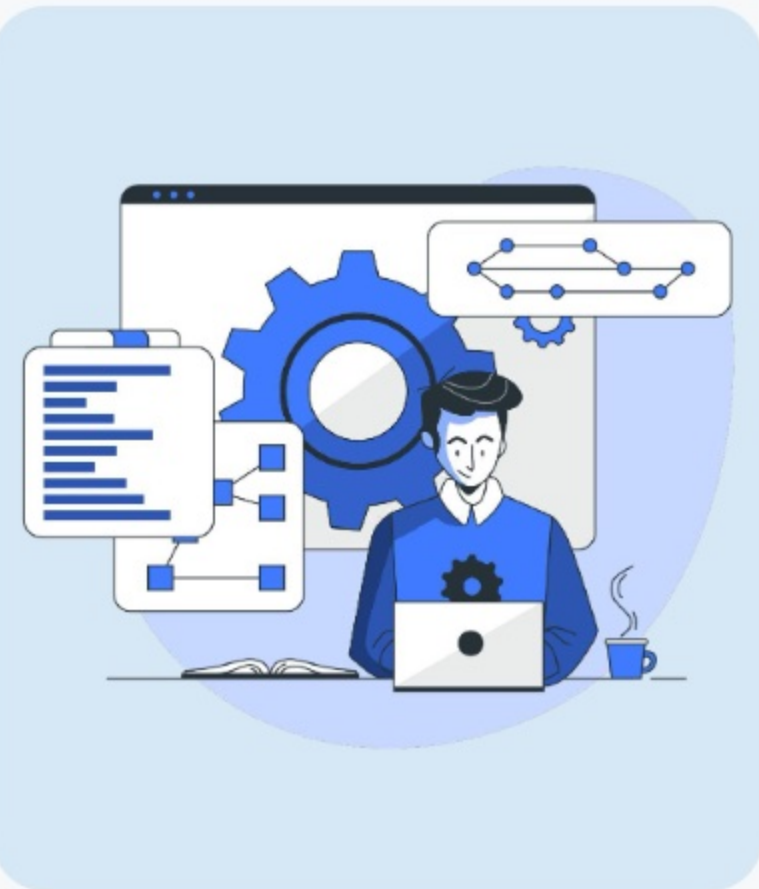
A sandbox serves as a **risk-free testing environment**, separate from the live production org, which enables **safe simulation of changes**.

❖ ACCURACY

Sandboxes **mirror the production org's tools and functionality** and **facilitate accurate testing and development** without risking live environment integrity.

❖ ASSESSMENT

Before making a change in a sandbox or directly in the live environment, the **nature of the change** and its **potential impacts** should be assessed.



Direct Changes in Production

Organizations typically have a **production org with live data** and **one or more sandbox environments** for development and testing, ensuring **changes do not affect live data** or **disrupt business operations**.

❖ CODING CHANGES

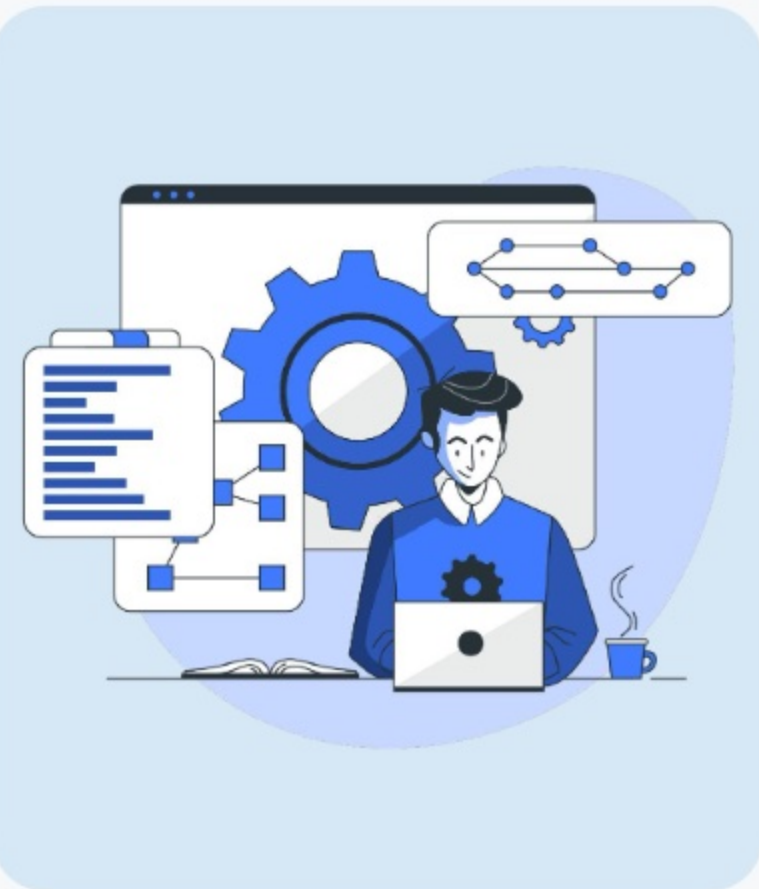
Coding changes **require a sandbox** since code cannot be written directly in a production org.

❖ HIGH IMPACT CHANGES

For changes that require **extensive testing** or **could impact system-wide operations**, such as new workflow steps or global feature implementations, sandboxes should be used for development and testing.

❖ LOW IMPACT CHANGES

Many low impact changes, such as **new custom fields**, **picklist values**, **reports**, and **dashboards**, can be made in production after taking precautionary steps to avoid any unforeseen changes.



Direct Changes in Production

Organizations typically have a **production org with live data** and **one or more sandbox environments** for development and testing, ensuring **changes do not affect live data** or **disrupt business operations**.

❖ DIRECT CHANGES

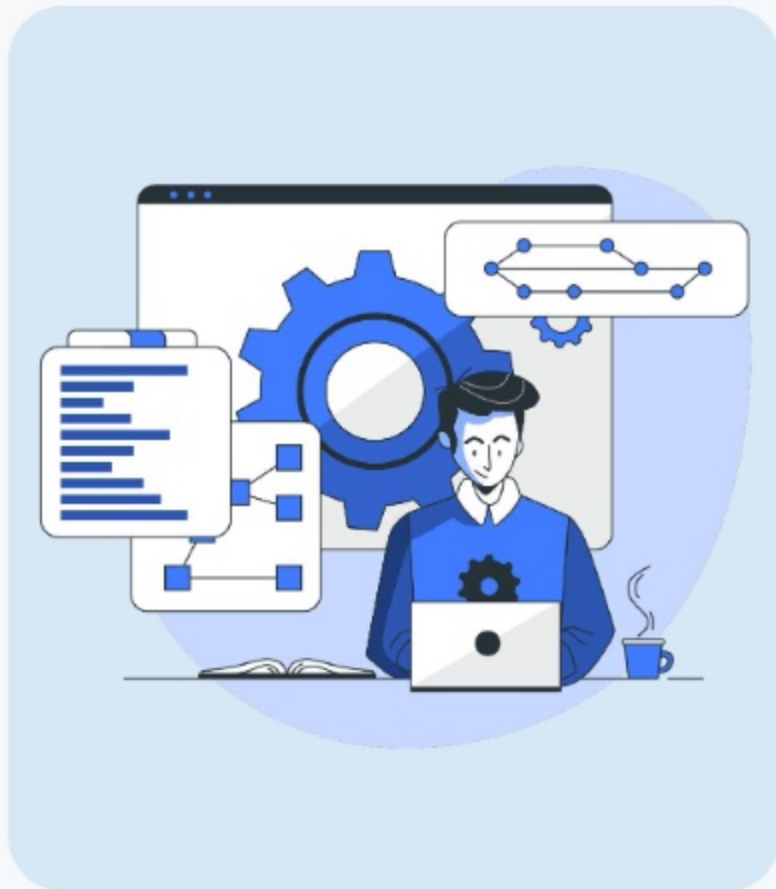
Direct changes in the production org should be approached with **caution**. **Steps** should be taken to **ensure adoption** and **avoid negative impacts**.

❖ DOUBLE-CHECKING

Thorough double-checking should be performed to make sure that the **change does not have any unwanted direct** or **indirect impact** and **no one else is using the change** for other purposes.

❖ CLEAR COMMUNICATION

Changes should be **clearly communicated to the team** before they are made. Users in the organization **should not be surprised** with changes.



Direct Changes in Production

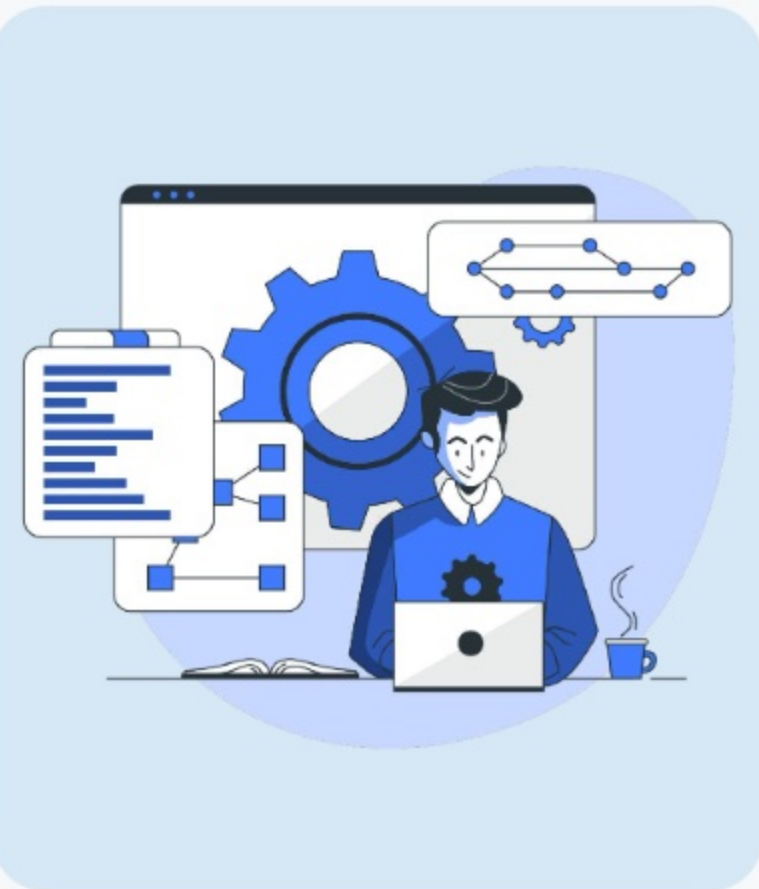
Organizations typically have a **production org with live data** and **one or more sandbox environments** for development and testing, ensuring **changes do not affect live data** or **disrupt business operations**.

❖ PHASED RELEASE

The change should be **validated** by other **admins**, **superusers**, or the **change requester** before it is released to the rest of the organization.

❖ TRAINING

User training can be provided to **get users on board** with the change, **explaining the change** and **how best to apply it**.



Implications of Direct Changes in Production

Before making a **direct change** in production, the **implications** should be considered.



ERRORS

Direct modifications **bypass critical testing**, leading to errors that are **harder to diagnose** and **rectify** without affecting users.



CONFLICTS

Changes might **conflict with other concurrent updates or integrations**, causing **unexpected behavior** or **system failures**.



RISKS

Regulatory and compliance risks may increase if changes impact **data security** or **user privacy** without undergoing rigorous validation.

Strategies for Direct Changes in Production

Certain **strategies** can be utilized for making **direct changes** in production.



HOTFIXES

A **dedicated process** can be established for **emergency hotfixes**, including criteria for what constitutes an emergency change warranting direct production updates.



REVIEW

Close collaboration should be fostered between **developers, QA team, and stakeholders** to ensure changes are thoroughly **reviewed and authorized**.



EXECUTION

Salesforce's **release management tools**, including deployment tools, and **sandbox refresh cycles** should be utilized to plan and execute changes methodically.

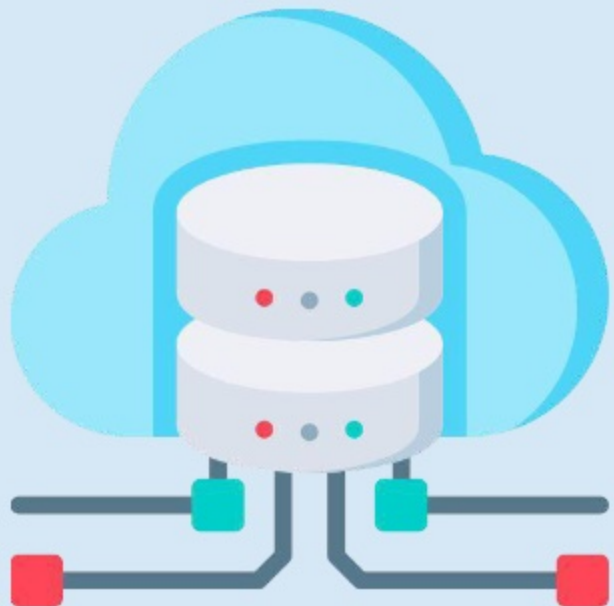
Business Scenario

Table of Contents



Business Scenario

Read the business scenario below and determine the most appropriate solutions for the company's requirements.



Cosmic Cloud Solutions, specializing in comprehensive cloud services, leverages Salesforce to streamline its operations and customer interactions. The company is facing critical issues in its production environment that are impacting service delivery and client satisfaction. These issues can cause significant brand damage and data inconsistencies. With an urgent need to resolve them without disrupting ongoing services, the company seeks the guidance of a Salesforce architect. The architect's role is pivotal in devising a strategy that includes immediate fixes in the production environment, ensuring minimal operational impact while maintaining the highest security and compliance standards. The architect must recommend a meticulous approach for resolving the issues while balancing the urgency of fixes with the need for thorough testing and stakeholder communication. The implications of making direct changes in production must be considered.

Requirements & Solutions

Table of Contents 

Requirement 1



SCENARIO

A developer has discovered a serious defect in production. Thousands of customer account records will be affected, resulting in incorrect data and causing major brand damage. A Salesforce consultant has suggested replacing the defective text field with a new picklist field directly in production. The page layout will also need to be modified accordingly. The architect needs to provide urgent assistance to resolve the issue.



Solution 1



SOLUTION

The architect should **collaborate closely with the consultant** to **review** and **implement** the changes carefully in the production environment. This pragmatic approach allows for real-time problem-solving and damage control, minimizing the impact on the brand while ensuring that changes are made thoughtfully and with oversight.

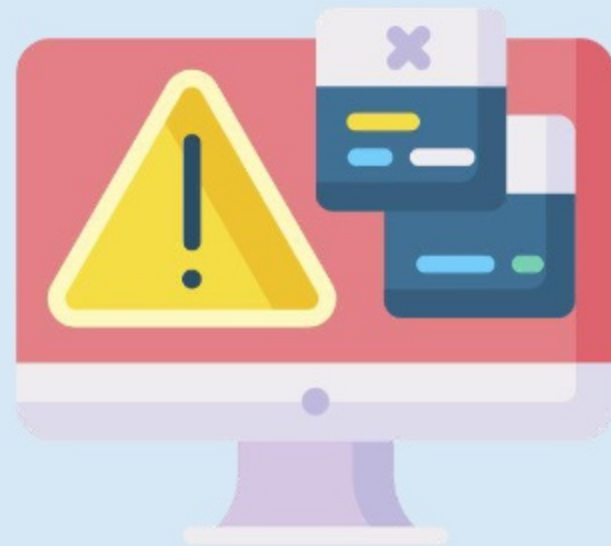


Requirement 2



SCENARIO

The company operates in a strict regulatory environment with dedicated teams for development, quality assurance, security, and release management. One of the Salesforce development teams is midway through a large enterprise project. It follows the agile development methodology and has four-week sprints. A severe production issue has been identified at the start of a sprint. The root cause is a bug in an Apex class deployed to production two months ago. Dozens of users use the production org regularly. The architect needs to recommend a suitable approach to address the bug.



Solution 2



SOLUTION

Since the serious production issue is tied to a bug in Apex code, understanding the scope and effect of the bug on existing data is crucial. **Investigating potential data impacts** would help assess the severity of the issue and effectively plan the necessary corrective actions. A **hotfix branch** should be created for the swift isolation and resolution of the bug without disrupting the current sprint's workflow. However, due to the strict regulatory environment, **stakeholder approval** should be obtained for implementing the hotfix to ensure compliance and transparency. Also, it is important to note that **Apex code cannot be written or updated directly in production**



Learn More

-  [Salesforce Sandbox – Managing Changes in Production](#)
-  [Salesforce Deployment Best Practices](#)
-  [The Difference between Salesforce Production & Sandbox](#)

